

Native Association: Confidence-Aware Human Perception in the Wild with a Foundation VLM

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Abstract. Extracting who is where, on which team, wearing which number from a broadcast frame is typically done by stitching a detector, an OCR engine, and classifiers together—and the stitching step swaps identities under occlusion. We make association *native* instead: a 0.77B vision–language model (Florence-2) is fine-tuned to emit all per-person attributes as one grammar-constrained sequence, with each attribute generated inside its owner’s block. Output is therefore schema-valid on every frame by construction, and no post-hoc binding step exists to attach a correctly read number to the wrong player: residual misassociation is pure perception error, $\approx 4\times$ rarer than zero-shot-prompted frontier APIs’ (0.057 vs. 0.21–0.24). On a frozen multi-sport test set, this single pass reaches 0.95 detection F1 (APIs: 0.65–0.75). A single extra forward pass yields a per-field confidence that supports a reject option (jersey precision $0.71 \rightarrow 0.96$ at half coverage) and routes a training-free zoom-and-re-read for small players. Surprisingly, once the grammar is learned, further parameter-efficient tuning yields no measurable gain under the adaptation configurations we test; the identical recipe on WIDER-Attribute reaches 93.1 mAP given-box, yields the first detection-coupled end-to-end results under its standard test protocol (84.5 mAP), and reproduces the same tuning result. In this regime, the gains live in the structure, not in added weights.

Keywords: Sports perception · Vision–language models · Grounded generation · Per-field confidence

1 Introduction

Dense, grounded, per-person metadata—every athlete’s box, team, jersey number, on-body text, and scene context—is a deployment problem before it is a modeling problem: it underpins sports analytics and archive-scale retrieval. Sports broadcast frames are a concrete instance of human-centered perception in the wild: motion blur, heavy inter-player occlusion, extreme scale variation (one athlete may fill the frame, a teammate <128 pixels), and small, rotated, warped typography.

Two families dominate deployment, and both fail at the same joint: association. *Disjoint cascades* chain a detector, an OCR engine, and classifiers, then

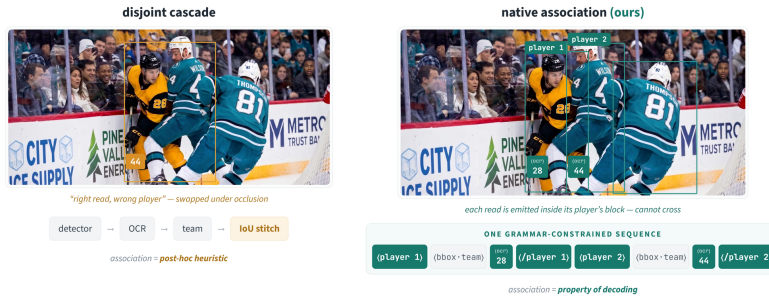


Fig. 1: Native association vs. post-hoc stitching. A disjoint cascade reads the number correctly but binds it to the wrong athlete via IoU stitching under occlusion (left); our model emits each read inside its owner’s `<player>` block of one grammar-constrained sequence, so the binding step does not exist (right). Quantified in Tab. 1 (jersey-digit AER) and Sec. 5.5.

stitch their outputs by an IoU heuristic; under occlusion the stitch attaches a correctly read number to the wrong athlete—even granted ground-truth boxes and one crop per athlete, cascades still misbind 15–18% of the jersey digits they read (supplementary). *Frontier vision-language APIs* offer strong open-world semantics but—on this task—weak grounding, unreliable output structure, no per-field confidence, and seconds of latency per frame (Sec. 5.5).

These failures live on three distinct axes, kept separate throughout: *syntactic validity* (does the output parse into the schema on every frame?), *visual grounding* (is every attribute anchored to a real, non-degenerate region?), and *semantic association* (is every attribute bound to the correct person?). Cascades fail chiefly the third axis (the stitch); the APIs degrade on all three.

We make association native instead, mapping one mechanism to each axis. A single 0.77B foundation VLM is fine-tuned to emit all per-person attributes as one grammar-constrained sequence, each athlete’s number and text generated *inside* that athlete’s block (Fig. 1). The finite-state grammar makes every frame schema-valid by construction (validity); in-block emission removes the post-hoc binding step, cutting the jersey-digit association error rate (AER: the fraction of correctly read numbers not bound to their true owner; Sec. 4) to about a quarter of the APIs’ (association); and, because grounding quality is pixel-bound for small players, a confidence-routed, training-free crop re-query re-reads exactly them (grounding), with one extra teacher-forced pass exposing the per-field confidence that routes it and supports a reject option (Sec. 5.4).

To test that this is a recipe rather than a lucky checkpoint, we transplant it unchanged—same schema, grammar, and frozen towers—to the public WIDER-Attribute benchmark [19] (Sec. 6). On both domains, further parameter-efficient tuning beyond the fine-tuned decoder moves the deployed metric by less than noise under the configurations we test (Sec. 5.3); in this regime the gains live in the structure, not in added weights. All systems share one frozen, fingerprinted test set and size-stratified analysis (Sec. 5.2).

Contributions.

- **A native-association single-pass architecture.** Grammar-constrained decoding emits every attribute *inside* its owner’s block, eliminating the post-hoc binding step— $\approx 4\times$ less misassociation than frontier APIs—and guaranteeing schema-valid output on every frame (Sec. 5.5).
- **Deployable per-field confidence.** One teacher-forced replay yields per-field confidence—a signal no baseline exposes—supporting a reject option (jersey precision $0.71 \rightarrow 0.96$ at half coverage) and routing a crop re-query.
- **No measurable gain from tuning beyond the grammar.** Once the grammar is learned, further parameter-efficient tuning beyond the fine-tuned decoder shows no measurable gain under the tested adaptation configurations, on both domains (Sec. 5.3); the same recipe reaches 93.1 mAP given-box and 84.5 mAP end-to-end on WIDER-Attribute (to our knowledge the first detection-coupled result), with checkpoint and recipe released (Sec. 6).

2 Related Work

Autoregressive spatial perception. Recasting localization as sequence generation is established—Pix2Seq emits boxes as discrete tokens [2], Kosmos-2 grounds phrases to regions [22], and Florence-2 unifies detection, captioning, and OCR in one prompt-conditioned decoder over a DaViT backbone [5, 30]—as is constrained decoding for entity linking [4], text-to-SQL [24], and general grammar-constrained LLMs [28]. Zero-shot, these AR models cannot serve as comparison rows: neither Pix2Seq nor Kosmos-2 emits a schema-typed, multi-field per-person record (team, jersey, on-body text), so no digit-association protocol exists to score them under; fine-tuning them on our corpus would change the backbone, not the mechanism under test. DETR-style object queries [1] bind attributes to instances natively but require trained task heads per attribute and offer no schema- or text-validity guarantee; generative grounded-NER frameworks [32] emit entity–region associations within one sequence, but over image–text pairs with detector-proposed regions and no validity guarantee, and multimodal constrained decoding has targeted single-entity outputs [31]. Our contribution is thus not any single component—structured serialization, constrained decoding, token-likelihood confidence, and crop re-querying are each established—but the composition in which the grammar *carries association*, not merely well-formedness: to our knowledge, we are the first to use a finite-state grammar to enforce association *and* schema validity by construction in multi-entity extraction from images alone.

Frontier VLMs and task-specific recognition. Production vision–language APIs (*e.g.*, Gemini [8]) can be prompted to return per-person JSON and, in schema-constrained modes, guarantee JSON well-formedness; no mode, however, guarantees grounded, non-degenerate geometry or exposes per-field confidence,

at seconds per frame. We evaluate the APIs free-form—their accuracy-best configuration [26]—and report first-attempt validity in Sec. 5.5; our decoder guarantees schema-valid, non-degenerate output with a usable confidence at $6.9\text{--}21\times$ lower latency. Person/pedestrian attribute recognition (PAR) targets the same per-person labeling but on *given* crops, from attribute-identity CNNs [20] to LLM-based recognizers [14] (no comparable end-to-end WIDER number); none predict attributes jointly with their grounding, as our single pass does.

Disjoint pipelines and confidence. The dominant deployed approach stitches specialists—a detector [16, 21], an OCR engine [6, 15, 18], and a team/attribute classifier—by post-hoc IoU [25]; each component can be strong in isolation, but the stitch is fragile under occlusion. SoccerNet jersey-number systems [10, 17] read numbers from tracklet crops with detection assumed—not directly comparable to our per-frame detect-and-bind protocol. Calibration [11] and reject-option classifiers [7] trade coverage for reliability but are rarely available from generative VLMs; we obtain per-field confidence from one teacher-forced replay, show it yields deployable reject options on both sports and WIDER-Attribute [19], and report which internal signals are uninformative.

3 Method

Our engine is a single foundation vision-language model (Florence-2-large, 0.77B parameters [30]) fine-tuned to map one image to one structured sequence that names every athlete and all of their attributes. Florence-2 couples a DaViT vision encoder to a BART-style encoder-decoder: visual embeddings and the embedded task prompt are fused by the transformer encoder, and our serialization is the decoder target. We freeze both encoders and fine-tune the decoder and shared token embeddings (details in the supplementary). The fine-tuning aligns the vocabulary; the guarantees come from the structure: a fixed-slot schema and finite-state grammar make association structural and schema validity guaranteed (Sec. 3.1), one replay pass exposes per-field confidence (Sec. 3.2), and an optional confidence-routed crop re-query recovers small players without training (Sec. 3.3). The training recipe is deliberately minimal (Sec. 3.4).

3.1 Unified Schema and Grounded Single-Pass Decoding

We expand the tokenizer with atomic structural tokens (`<sport>`, `<stype>`, `<player>`, `<bbox>`, `<team>` and its four flags, `<ocr>`, `<gdesc>`) and serialize the full per-frame annotation into one target sequence whose emission order is enforced by a prefix-constrained finite-state grammar—the serialization below is the only shape a decoded sequence can take (visualized in the supplementary):

```

      <sport>{s} <stype>{t}
( <player> <bbox> <loc>×4 </bbox> <team>{flag}
  ( <ocr>{text} [<loc>×8] ) *
    </player> ) *
  <gdesc>{desc} </s>

```

Continuous coordinates are quantized to 1000 location bins and emitted as `<loc_*>` tokens; the implemented grammar bounds the starred repetitions (12 players, 17 OCR items per player). Player blocks are emitted largest-first (bbox area, descending) and OCR items largest-quad-first—a deterministic canonical order; under the grammar we observe none of the repetition or early-termination failures that motivated Pix2Seq’s randomized ordering and sequence augmentation [2] (removing the grammar, by contrast, collapses decoding entirely; Sec. 5.3). The eight per-OCR quad tokens (bracketed above) are optional: our headline configuration omits them, since quad supervision is net-negative for reading (−3.8 jersey-F1 for +0.4 detection-F1; ablation in Sec. 5.3); the quad variant is retained for downstream tasks needing per-word geometry.

Two properties follow *by construction*, not by training. First, **native association**: every OCR string (and, when emitted, its coordinates) is generated inside exactly one `<player>...</player>` block, so text–player binding is read off the sequence rather than reconstructed by a post-hoc IoU stitch. This does not make the binding infallible: a residual *perception* error remains, in which the model reads a neighbour’s number into the wrong block; we measure it as the association error rate (AER; defined in Sec. 4)—0.057 single-pass, $\approx 4\times$ below the frontier APIs’ 0.21–0.24, whose grounding is itself post-hoc. Second, **guaranteed validity**: a prefix-allowed-tokens function admits only grammar-legal continuations at each step, so every decoded sequence parses into the schema—validity 1.0 on every frame, with no post-hoc repair and no reliance on the model having “learned” to be well-formed. The result is a *typed serialization* whose single-token keys avoid literal JSON’s 3–5 BPE tokens per key and its malformability.

3.2 Per-Field Confidence from a Single Teacher-Forced Replay

After constrained generation, one teacher-forced forward pass (fp32) over the generated tokens recovers the per-token log-probabilities—no second generation, no auxiliary head (≈ 150 ms, $\approx 10\%$ of generation; supplementary). Generation-time scores are grammar-masked, so we replay *unmasked*: confidences are full-vocabulary probabilities—the model’s raw beliefs, not values renormalized over the few grammar-legal tokens. For any contiguous field $F = (y_{t_1}, \dots, y_{t_k})$ we define its confidence as the geometric mean of its per-token probabilities,

$$c(F) = \exp\left(\frac{1}{k} \sum_{i=1}^k \log P(y_{t_i} \mid y_{<t_i}, X)\right), \quad (1)$$

the length-normalized likelihood of the field given the image X (for one–two-token jersey numbers, close to a single-token probability; an ablation of the aggregator—geometric vs. arithmetic mean vs. min-probability over the same replay logits—is in the supplementary, and finds the choice immaterial for these short fields). Applied to the text tokens of an OCR item, Eq. (1) gives its *text_conf*; a player’s *jersey_conf* is the *text_conf* of its first digit-only OCR item (an emission-order tie-break; largest-quad when quads are emitted). From

the same logits we also read a structural *loc_conf* (geometric mean over the four box-location tokens) and an *objectness* margin ($P(\langle \text{player} \rangle)$ vs. $P(\langle \text{gdesc} \rangle)$, renormalized over that two-way choice at the block-opening step). Sec. 5 shows *jersey_conf* separates correct from incorrect reads, supporting a reject option, while the structural signals saturate and serve only to route the re-query—a negative result we report explicitly. Cascade components expose per-component scores (detector objectness, OCR recognition confidence), but no unified confidence over the associated player–number pair; closed VLM APIs expose none by default.

3.3 Training-Free Crop Re-Query

Small and heavily occluded athletes are the dominant error mode (Sec. 5); we address them at inference time only, with a zoom-and-re-read of the selected players. From the first-pass prediction we select players that are small (bottom quartile by predicted box area) or uncertain (bottom quartile by *jersey_conf*), crop a padded region around each, and re-run the *same* model on the crop under the same grammar capped at one player. The re-read attributes replace the first-pass ones in place—boxes are never re-localized, so no duplication can arise—and each emitted player is tagged with its source (frame or crop). The trade is explicit: re-reading recovers small players (+0.14 tiny-bucket jersey F1) at a small association cost ($0.057 \rightarrow 0.068$ AER; Sec. 5.5), since a crop that straddles a pile-up can bind a neighbour’s number. No test label informs the selection; the padding fraction is chosen on validation by jersey-number F1, and crops decode as short single-player sequences, so the average cost stays modest (1.93 passes/image; Sec. 5). The confidence-gated flow is diagrammed in the supplementary.

3.4 Training Recipe

Training is intentionally plain. We warm-start the new token rows with the mean embedding of descriptive sub-words (*e.g.*, $\langle \text{player} \rangle \leftarrow \text{“player”}$), freeze both encoders, and fine-tune the decoder plus the shared token embeddings with a uniform cross-entropy loss; the grammar plays no role in the teacher-forced loss—validation decoding for checkpoint selection is grammar-constrained from the first epoch. Checkpoints are selected on a held-out split by an aggregate *val_score* (unweighted mean of five task metrics; components in the supplementary); differences below +0.005 on this 0–1 composite are treated as noise (single-seed runs; see limitations), and the frozen, fingerprinted test set is touched once. We use no multi-stage curriculum, loss weighting, or adapters in the main model; the adapter arm is quantified as a null in Sec. 5.3 (full recipe and hyper-parameters in the supplementary).

4 Dataset and Evaluation Protocol

Data and splits. Our sports corpus is broadcast frames from multiple sports, deduplicated (perceptual hash) and quality-filtered. Each frame carries per-person annotations—axis-aligned box, team $\in \{\text{A, B, other, unknown}\}$, jersey number (digit string, possibly empty), on-body OCR items (text, optional quadrilateral)—plus sport type, scene type, and a free-form description. The corpus spans five sports—hockey, soccer, gridiron football, tennis, and basketball, plus a small *other* bucket (per-sport frame and player counts in the supplementary)—is dominated by in-game footage, and was labeled by an LLM-assisted pre-annotation pass corrected by human annotators in CVAT over iterative review cycles. One frame-level partition gives train/val/test = 4,584/577/567 frames (10,408/1,325/1,266 players; 33,633/4,185/4,046 OCR items). The split is keyed on each frame’s perceptual hash (stratified by sport and athlete count), so identical or near-duplicate frames cannot straddle splits. Provenance metadata shows 5,623 of the 5,728 frames are unique-URL still images; 105 come from 22 source videos, of which three span splits: exactly 4/567 test and 6/577 validation frames share a video with training. Match identity across *distinct* still assets is untracked, so photographs of one match may fall in different splits; we read our numbers as within-domain and leave a match-disjoint split to future work. Validation drives all model and hyper-parameter selection; the test split is frozen and decoded once per model variant—every subsequent analysis is an offline re-score of those cached predictions—under a fingerprint guard: each evaluation recomputes the frame count, frame-set hash, and annotation-content hash and aborts on mismatch (provenance in the supplementary).

Metrics. One evaluation suite scores every system—ours, APIs, cascades—with identical matching. *Detection*: Hungarian assignment at $\text{IoU} \geq 0.5$ (P/R/F1). *Team*: exact accuracy (which scores the annotator’s A/B convention) and a permutation-invariant purity—per-frame Hungarian assignment between predicted and ground-truth team labels, micro-averaged over matched players. *Jersey* (digits only): the any-OCR rule—a ground-truth number is correct if it appears as any emitted digit token for the matched player (strict exact-string variant in Sec. 5.3). *OCR text*: fuzzy Hungarian at normalized Levenshtein $\tau = 0.7$, pooled per frame and deliberately association-blind—binding errors are charged to AER, not text F1. *Output validity*: parse-success, empty-output, schema-valid rates. *Confidence* (Sec. 5.4): AUROC, ECE, precision at fixed coverage. Scene metrics and full definitions are in the supplementary. *Association*: AER is the fraction of correctly-read jersey digits not bound to their true owner—covering both binding to a wrong matched player and grounding to no valid box (unmatched or degenerate)—with the same matcher for every system. We report the digit-only rate because team text is identical across teammates, so its owner is ill-defined; we computed the all-text rate first, and its full decomposition (all systems, all error classes) is in the supplementary. The choice matters: under

the all-text rate the oracle-boxed cascades attain *lower* AER than our single pass (0.057–0.070 vs. 0.085), driven by exactly those owner-ambiguous team-text items; the digit-only ranking is specific to it. For oracle-boxed cascades the denominator spans only the numbers they read (Tab. 1[†]).

Size stratification. To expose the small-player regime, IoU-matched players are bucketed into quartiles by ground-truth box area; detection recall, jersey F1, and OCR-text F1 are reported per bucket (Sec. 5.2).

Latency and cost. Accuracy metrics are computed in `fp32`; self-hosted latency is an `fp16` pass over the full test set (the grammar ablation in the supplementary reports `fp32`); API latencies are end-to-end wall-clock and not strictly cross-comparable (details in the supplementary).

Reproducibility. We will release: the evaluation suite, the finite-state grammar and inference harness, the verbatim API prompts and retry policy, the per-system prediction caches behind every number in Sec. 5, the WIDER-Attribute checkpoint, and a recipe that reproduces the training and full pipeline on any similarly-annotated dataset. The sports corpus and checkpoint are proprietary and not released: the WIDER-Attribute results are thus reproducible end-to-end; the sports numbers are re-scorable from the released caches, whose provenance is pinned by the frozen-test fingerprint.

5 Experiments

5.1 Main Comparison

We benchmark against two production vision–language APIs (Gemini-2.5/3.1-pro) and two disjoint tiers—an out-of-the-box Florence-2 cascade and a specialist detector–OCR–classifier pipeline—on the frozen 567-frame multi-sport test set, all scored through the *same* suite with identical matching (Hungarian, $\text{IoU} \geq 0.5$) so the columns are directly comparable (Tab. 1).

Protocol notes for Tab. 1. Jersey metrics use the any-OCR digit rule, with the strict exact-digit variant alongside (Sec. 4). Gemini-3.1-pro is its strongest (medium-resolution) configuration; the *specialist cascade* is a PP-OCRv5 reader, and the two cascades share one team classifier. Match coverage is the fraction of GT players IoU-matched (equals detection recall). First-attempt parse failure counts first-try invalid JSON: other API cells are post-retry, ours is zero by construction, and Gemini-2.5-pro logged no retry telemetry. Latency and cost follow Sec. 4: `fp16` self-hosted (\$1.5–2/hr on-demand A100) vs. wall-clock list-price for the APIs—not cross-comparable.

Three patterns summarize Tab. 1. *Grounding is the chasm*: we detect at F1 0.95 where the APIs reach 0.65–0.75, and their reading deficit disappears once detection is conditioned away (given-detected jersey recall: APIs 0.82–0.83, ours

Table 1: Main comparison on the frozen 567-frame test set (1,266 players, 4,046 OCR items); all systems scored by the identical suite (metric rules in Sec. 4, protocol notes in Sec. 5.1). *Ours* (E1b) = single pass; *+re-query* (E3) = the training-free crop re-query (Sec. 3.3); these labels name the two configurations throughout. Our AER is statistically tied with the oracle-boxed cascades[†], and the single-pass jersey-F1 lead over Gemini-3.1 is within CI (Sec. 5.5). [†]Oracle GT boxes—generous to the cascades; their AER spans only the 32–37% of numbers they read (supplementary). [‡]Reading+team+stitch only, **fp32**, oracle boxes—no detector was run, so this latency (and its cost) is not end-to-end. All systems run one pass per image except *+re-query* (1.93 on average). Bold marks the best directly-comparable cell per row; oracle-assisted ([†]), non-comparable ([‡]), and inapplicable (–) cells are excluded. ↓ = lower is better; – = not applicable.

Metric	Ours (E1b)	Ours +re-query	Gemini 2.5-pro	Gemini 3.1-pro	Specialist cascade [†]	Florence-2 cascade [†]
Detection F1	0.950	0.950	0.652	0.745	GT [†]	GT [†]
Team acc. (exact)	0.809	0.809	0.838	0.880	0.558	0.558
Team purity (perm-inv)	0.952	0.952	0.974	0.994	0.764	0.764
Match coverage	0.950	0.950	0.704	0.772	GT [†]	GT [†]
Jersey F1	0.757	0.792	0.704	0.737	0.460	0.520
Jersey F1 (strict)	0.695	0.733	0.630	0.673	0.407	0.478
Jersey P	0.755	0.786	0.829	0.860	0.809	0.852
Jersey R	0.759	0.799	0.612	0.645	0.321	0.374
OCR-text F1	0.584	0.602	0.476	0.502	0.283	0.348
AER (jersey digits)↓	0.057	0.068	0.242	0.214	0.075 [†]	0.060 [†]
First-attempt parse fail. (%)↓	0.0	0.0	–	10.1	0.0	0.0
Schema valid (%)	100	100	–	–	–	–
Latency (ms/frame)↓	1224	1274	26009	8426	89 [‡]	708 [‡]
Cost (\$/1k)↓	~0.5–0.7 (A100)		9.11	7.11	~0.05 [‡]	~0.4 [‡]

0.78; supplementary)—it is a detection deficit. *Precision-by-abstention*: every baseline family reads only the easy players (coverage 0.70–0.77; oracle cascades read 32–37% of numbers), which inflates their precision; we read everyone and can trade coverage for precision explicitly (Sec. 5.4). At each API’s own recall (0.65 / 0.61) our reject option reaches jersey precision 0.91 / 0.92 (E1b) vs. their 0.86 / 0.83 (supplementary): we dominate the precision–recall trade and, unlike the APIs, can set this operating point explicitly. *Association*: our misbinding rate is $\approx 4\times$ lower than the APIs’ (Sec. 5.5); a qualitative comparison on a crowded frame is in the supplementary. One scoping note: Tab. 1 compares the complete fine-tuned system with zero-shot baselines—deployment-level superiority, not a per-component decomposition; the oracle-boxed cascade rows isolate the stitching step (Sec. 5.5), and a domain-trained cascade, which would separate native association from domain adaptation, is future work (Sec. 7).

5.2 Performance by Player Size

Reading is size-bound, and detection is where the APIs lose small players (full size table and plot in the supplementary). For every system, jersey and OCR-text F1 rise with player size—monotonically except in two cells—so the residual error budget lives in the small-player regime. The failure modes differ, however:

our tiny-bucket detection recall is 0.85 vs. Gemini-3.1’s 0.56—an API that never localizes a small athlete cannot read it—whereas our own tiny-bucket shortfall is a *reading* deficit on *detected* players. The training-free crop re-query (Sec. 3.3) targets exactly this bucket, lifting tiny-bucket jersey F1 $0.61 \rightarrow 0.75$ (63 reads gained, 26 lost, net +37) while leaving larger buckets unchanged.

5.3 Ablations

Our central observation is that, once the grammar is learned, accuracy is governed by inference-time and schema *structure*, not by additional parameter tuning. Parameter-efficient adapters (LoRA and relatives [13]) are the default recipe for squeezing extra accuracy from a frozen backbone, so we include a LoRA-target ladder as an ablation and find it null—a negative result that motivates this framing rather than a new tuning method. Three ablations on the frozen test (all at the pre-registered `best_model_score` checkpoint, jersey scored by the strict exact-digit rule for internal consistency) support this.

Quad supervision (a structural choice). Emitting the eight per-OCR quadrilateral tokens is optional (Sec. 3.1). Dropping them (our headline E1b) *improves* reading and speed while costing almost nothing on detection (full table in the supplementary): +3.0 jersey-F1 under the strict exact-digit rule (+3.8 under the any-OCR digit-token rule used in Tab. 1), +1.6 OCR-text-F1, +3.7 team purity, and a 28% latency reduction ($1710 \rightarrow 1224$ ms), against a -0.4 detection-F1 change. The extra geometry supervision spends decoder capacity that is better used for reading; we keep the quad variant only for the ablation and for downstream tasks that need per-word boxes.

The LoRA ladder shows no measurable gain. Starting from the trained decoder we add LoRA adapters of increasing capacity and re-select on validation: no adapter variant (decoder V / vision / decoder QVO / decoder QKVO, up to 3.15M trainable parameters) moves the aggregate by more than +0.0011, roughly $5\times$ under the +0.005 floor (Sec. 3.4; table in the supplementary). These are single-seed runs gated by a pre-registered noise floor, so the claim is scoped accordingly: no measurable gain was observed under the tested adaptation configurations, not that further tuning is universally ineffective. For contrast, the single *structural* decision of dropping quads (above) is worth +0.008 aggregate—roughly $7\times$ the best tuning delta—in this regime (frozen towers, a 0.77B backbone, and our data scale), structural choices rather than added capacity move the metric, and the pattern replicates on WIDER-Attribute (analogous tuning delta +0.02 mAP, Sec. 6).

The grammar constraint is load-bearing. In both domains, decoding the identical `best_model_score` checkpoint with the same greedy configuration but the finite-state constraint disabled collapses generation into an unbounded start-token loop: the decoder never emits a content token, hits the 1024-token cap, and

yields zero parseable persons (detection and jersey F1 0.000, empty-output rate 1.000, $7\times$ the latency)—on WIDER-Attribute exactly as on sports (full table and mechanism in the supplementary). The grammar’s first-token mask forbids this continuation, so the constraint both guarantees the schema *and* terminates generation (on sports, reducing wall-clock by 7,580 ms/frame, -85.6%). Read narrowly: this ablation isolates the grammar’s contribution to *decoding stability* and *format validity* only—no content token is ever emitted, so it says nothing about perception accuracy and we do not interpret it as a detection or recognition gain. Nor do we claim the loop is irrecoverable by other decode-time heuristics; the grammar removes it while *simultaneously* guaranteeing schema validity by construction, which ad-hoc anti-repetition fixes do not.

5.4 Confidence and the Reject Option

The single teacher-forced replay (Sec. 3.2) yields a per-player *jersey_conf* at no extra generation cost that separates correct from incorrect reads well (AUROC **0.879** E1b, 0.831 after crop re-query E3; team-margin AUROC 0.84; ROC and distributions in the supplementary). Because it is a length-normalized likelihood in $[0, 1]$, ranking players by it and abstaining on the least-confident trades coverage for precision (risk-coverage curve and operating points in the supplementary): E1b reads at precision 0.71 at full coverage, 0.87 at the top 70%, and 0.96 at the top half—a knob neither the cascades nor the closed APIs offer, with the E1b and E3 curves crossing near full coverage. Crucially this is opt-in—our headline numbers are at full coverage, so the reject option adds precision without quietly dropping recall to obtain the Tab. 1 figures—and it relies only on the *ordering* of *jersey_conf*, not its calibration (the raw likelihood is over-confident, ECE 0.12–0.15 across E1b–E3, removable with one temperature if absolute probabilities were needed).

Negative findings. Not every signal is informative: the structural *loc_conf* (AUROC 0.507 E1b, 0.573 E3) and the objectness margin—a renormalized two-way choice by construction, consistent with its saturation—are near-random correctness predictors on both domains, useful only as re-query routers (Sec. 3.3), never filters (detail in the supplementary).

5.5 Robustness and Statistical Reliability

Reliability, association, and significance. Grammar-constrained decoding parses on all 567 frames; the APIs carry no such guarantee (first-attempt failures in Tab. 1; full retry and empty-output accounting in the supplementary). The damaging failure for a grounding task is *invalid geometry*: Gemini-3.1-pro emits a degenerate (zero-area) bounding box¹ on 20.3% of its predicted players—at

¹ Zero width or height after normalization ($w \leq 0.002$ or $h \leq 0.002$); such a box has IoU = 0 with any ground-truth player and can be matched to no one.

least one on 22.2% of frames—and on 177 of those players the jersey text was *right* but grounded to no athlete, so it cannot be credited: reading without grounding, exactly the failure in-block emission removes. Consistently, 90–97% of API misbindings are such orphaned reads, whereas ours are contact-occlusion swaps between overlapping players (decomposition in the supplementary); and our fine-tuned model reads better in context (OCR-text F1 0.584–0.602, own detection, whole frame) than either specialist engine granted oracle per-player crops (Tab. 1 is the full-frame tier; Florence-2 OOTB 0.437, PP-OCRv5 0.357). A paired frame-level bootstrap (10,000 resamples of the 567 frames, the same resample scoring every system; full table in the supplementary) makes the headline gaps significant—detection F1 +0.22 to +0.30, jersey-digit AER +0.16 to +0.18, OCR-text F1 +0.09 to +0.11; every interval excludes 0—and we are equally explicit where it does not: our jersey-F1 edge over the strongest API is +0.02 $[-0.02, +0.06]$, and our digit-AER is statistically indistinguishable from the oracle-boxed cascades’ (+0.00 to +0.02, CIs spanning 0)—expected, since they are handed ground-truth boxes yet read only a third of the numbers.

6 Cross-Domain Transplant: WIDER-Attribute

To test whether our results reflect a general recipe rather than one dataset, we apply the same components—schema-token expansion, grammar-constrained decoding, frozen towers, teacher-forced confidence readout, and a crop second pass (applied here as late fusion over all persons rather than confidence-routed re-query)—to a second, public, human-centered benchmark: human attribute recognition in unconstrained event scenes on WIDER-Attribute [19] (people across 30 in-the-wild event categories—parades, ceremonies, meetings, rescues—rather than street pedestrians; test split: 6 918 images, 29 177 person instances, 14 binary attributes). The only change is the slot vocabulary (team/jersey/scene become 14 attribute flags); we fine-tune *only* on the WIDER training split—this is not zero-shot transfer of the sports weights, but a re-instantiation of the same procedure. Ground-truth boxes are provided at inference so the number is comparable to the attribute literature, which assumes given boxes.

Result. The model reaches **93.1 mAP** given-box (late fusion of a whole-image and a per-person-crop pass: scene 89.9, crop 91.9, fusion 93.1), at or above a decade of task-specific WIDER specialists (Tab. 2), with a single set of weights serving every inference protocol. Because our detector also localizes people, we additionally report a *fully end-to-end* number—detect, crop, and classify without ground-truth boxes—of **84.5 mAP**, to our knowledge the first detection-coupled result—no ground-truth boxes at any stage—under the standard WIDER-Attribute test protocol; the gap to the given-box number is dominated by an annotation-sparsity artifact (the detector fires on background people the annotators skipped: unmatched predictions have median box area 0.023 and 69.5% arise on images with more predictions than labels), so we treat the end-to-end figure as a lower bound and the given-box figure as the fair comparison

Table 2: WIDER-Attribute in context (Sec. 6). Prior methods are *given-box* only. mAP is the historical WIDER metric; mA (mean per-attribute accuracy) is common in recent PAR work. **mAP and mA are different metrics and are not comparable across blocks.** The table is for context, not a common-protocol ranking.

Method	Metric	Score
<i>mAP era (given-box)</i>		
R*CNN [9]	mAP	80.5
DHC [19]	mAP	81.3
SRN [33]	mAP	86.2
DIAA [23]	mAP	86.4
Da-HAR [29]	mAP	87.3
VAC [12]	mAP	87.5
<i>mA era (given-box)</i>		
VTB [3]	mA	88.2
PromptPAR [27]	mA	92.0
<i>Ours (this work)</i>		
Ours (given-box)	mAP	93.1
Ours (end-to-end, first reported)	mAP	84.5

point. The scope of this transplant is stated explicitly: WIDER’s protocol tests the *recipe*’s generality—schema, grammar, confidence readout, second pass—not the text–person association claim itself; the end-to-end variant exercises grounding but not text binding.

The same phenomenology. All three sports findings replicate. *No measurable tuning gain:* extra adaptation moves the deployed given-box metric by < 0.1 point ($+0.02$ mAP), mirroring the LoRA ladder (Sec. 5.3); per-attribute threshold tuning is likewise unhelpful (-0.8 mA). *Usable confidence:* the per-field probabilities are calibrated here (ECE 0.028 after fusion) and abstaining on the least-confident 10% of $\approx 409k$ (person, attribute) decisions cuts error from 4.1% to 1.5%. *Routing, not filtering:* `loc_conf` routes the re-query benefit to low-confidence detections yet is a weak filter, and objectness saturates, exactly as on sports (Sec. 5.4; a bonus visibility-detection property of the uncertainty token is described in the supplementary).

7 Conclusion

We presented a single-pass perception engine that makes structure intrinsic to generation: a 0.77B VLM fine-tuned to emit every per-person attribute inside a grammar-constrained sequence, so association is a property of decoding, not a post-hoc stitch. It is schema-valid on every frame, leads detection (0.95 F1), reads at full coverage where the APIs abstain, at far lower latency, and cuts jersey-digit association error $\approx 4\times$ below theirs. In our regime (frozen towers, a 0.77B backbone, single seed), further parameter-efficient tuning showed no measurable gain under the tested configurations—once the grammar is learned,

accuracy here is governed by structure—and the identical recipe on WIDER-Attribute (93.1 mAP given-box) reproduces it: a property of the recipe, not a lucky checkpoint.

Limitations. Accuracy on small players is bounded by legible pixels: the 1000-bin coordinate quantization and finite decoder resolution limit tiny-text reading, and quality is detection-recall-bound, so residual error concentrates in the small-athlete bucket (Sec. 5.2), mitigated but not eliminated by the crop re-query. Confidence thresholds and re-query padding are validation-selected, not learned, and the raw likelihood is over-confident (ECE 0.12–0.15). Our cascade baselines are zero-shot / oracle-assisted (ground-truth boxes isolate stitching from detector quality); the twelve-person cap is non-binding on our data, and the grammar in practice *reduces* wall-clock $\approx 7\times$ (Sec. 5.3). All runs use a single seed, with tuning deltas gated by the pre-registered noise floor; residual failures are in the supplementary. Confidence-gated override of pass 1, temperature scaling, and a domain-trained cascade comparison are natural next steps, each validation-decidable and left untuned here.

Ethics and intended use. The system reads jersey numbers and clusters athletes into *relative* per-frame teams in professional broadcast footage only—no identity or face recognition—for sports analytics and archive indexing, not surveillance.

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